

Nafee Karim

Nafkarim007@gmail.com | 832-705-6838 | linkedin.com/in/nafkarim | U.S. Citizen

EDUCATION

The University of Texas at Austin

Austin, TX

B.S. Mechanical Engineering, Minor in Business; GPA: 3.4

May 2026

- Relevant Coursework: Thermodynamics, Fluid Dynamics, Materials Engineering, Mechatronics, Heat Transfer, Statistics, Mechanics of Solids, Dynamics, Programming, Machine Elements, Operations Research Models, Simulation Modeling, Data Science, Finance, Numerical Methods, Business Law, Management

WORK EXPERIENCE

IBM

2026 (Incoming)

Associate Business Transformation Consultant — ServiceNow

Samsung Semiconductor

May 2025 – Aug 2025

Systems and Process Engineering Intern

Austin, TX

- Led consolidation of 15+ fragmented data sources into a unified dataset covering thousands of semiconductor tools site-wide, enabling comprehensive operational visibility and forming the data foundation for predictive maintenance planning across critical fabrication systems.
- Owned product implementation and migration of CMMS platform, defining product requirements, roadmap milestones, dependencies, and acceptance criteria; managed cross-team migration of maintenance and tool-monitoring workflows, owning timelines, risk/issue trackers, and status communications to keep equipment readiness milestones on track while minimizing disruption to live wafer starts.
- Standardized spare-parts governance across 6 engineering teams by consolidating thousands of part codes and 250+ preventive maintenance (PM) plans; built Spotfire dashboards mapping PM plans to parts, tracking usage and lead-time trends to forecast demand and reduce duplicate reorders and surprise stockouts.
- Analyzed 100+ reactive maintenance events using Pareto analysis and FMEA to identify 25+ recurring breakdown drivers; partnered with equipment engineers to design targeted preventive maintenance plans aimed at reducing unscheduled downtime and improving Mean Time Between Failures (MTBF).
- Partnered with Quality Engineers on 8D-style root-cause investigations for deviations and near-miss events, coordinating with vendors, planners, and tool owners to isolate failure drivers, define corrective actions, and reinforce SOP adherence to mitigate safety and fab yield risk.
- Performed CAD modeling and CFD analysis of data server enclosures cooled by Liebert CW CRAC units; evaluated airflow and thermal performance to recommend ceiling height reduction, enhancing HVAC efficiency and mitigating heat stratification in high-ceiling server environments.

Qorvo

May 2024 – Aug 2024

Process Engineering Intern

Richardson, TX

- Led launch of an automated X-ray packaging inspection workflow for a high-demand RF device product generating over \$150M in revenue; reduced unit inspection cost from \$1.30 to \$0.25 (81% reduction) while significantly cutting manual labor time and laying the groundwork for a fully automated inspection process.
- Diagnosed plasma tool yield excursions through cross-functional data analysis spanning equipment logs, process records, and test results; identified chamber inconsistencies and tuned equipment settings to improve etch rate uniformity, resulting in a 1–2% yield improvement across affected tools.
- Standardized data-capture protocols for post-rework verification to ensure process repairs and equipment tweaks were consistently logged and traceable for future failure investigations; proactively managed route changes from new die attach applications to prevent production delays and clear a significant backlog of PFD updates.

Black Diamond Networks

Sept 2025 – Dec 2025

Associate Account Executive Intern

Austin, TX

- Built and managed a pipeline of 150+ prospective accounts by engaging 100+ hiring managers weekly through structured outreach campaigns, converting 10–15% into active client engagements via disciplined needs discovery, relationship development, and consistent follow-up cadences.
- Executed search delivery alongside account executives and recruiting teams across 100+ concurrent client searches, coordinating 5–10 candidate submissions per requisition and leveraging CRM tooling and LinkedIn sourcing across 50+ target accounts per week.
- Synthesized intake calls and discovery conversations into concise role requirement summaries and structured status updates for account managers.

Texas Guadalupe (Student Organization)

Feb 2023 – May 2023

Braking / Manufacturing Engineer

Austin, TX

- Contributed to development of the pod's primary braking mechanism for a hyperloop competition vehicle, leveraging SolidWorks for detailed system design and conducting FEA simulations to validate structural integrity and performance across operating and worst-case loading scenarios.
- Developed hands-on expertise in industrial machining including manual mills, CNC mills, and lathes, translating engineering designs into practical manufacturing processes.

RESEARCH EXPERIENCE

Kovar Research Group

Sept 2024 – Present

Undergraduate Research Assistant

Austin, TX

- Conducting research on additive manufacturing of ceramics focused on enhancing material density and thickness; responsible for producing Aluminum and Zirconia pellets processed through high-temperature sintering and Selective Laser Flash Sintering (SLFS), followed by systematic measurements to evaluate changes in material properties and performance.
- Responsible for setting up, calibrating, and maintaining laboratory equipment including SLFS and tensile testing machines, ensuring consistent experimental conditions and data quality across research sessions.

Hutter Research Group

Sept 2023 – Dec 2023

Undergraduate Research Assistant

Austin, TX

- Collaborated with a graduate student on testing of a prototype medical device designed to analyze bio-fluids within the brain; designed and 3D printed a custom container for controlled solution evaporation on a heat plate and performed Ansys thermal and fluid simulations to optimize the process and improve testing accuracy.

PROJECTS

RecommendMe — Recommendation Score Prediction Model

Data Science Project

Collaborative Filtering + Cold-Start ML Model

2025 – 2026

- Partnered with startup RecommendMe (platform connecting recommendation-letter requesters with writers) to build a model predicting the 1–5 star rating any recommender would give any requestee; trained an SVD collaborative-filtering model (Surprise library, 40 latent factors, 100 epochs) on 500 rating edges across 50 users.
- Evaluated model with MAE, RMSE, R^2 , exact rounded-star accuracy, and within- ± 1 -star accuracy under an 80/20 train-test split and 5-fold cross-validation; built a classifier block routing each new recommender-requestee pair to SVD, new-requester, new-recommender, or cold-start prediction paths.
- Solved the cold-start problem for users with no rating history by training a Gradient Boosting regressor on TF-IDF features (1–2 grams, 1,000 max features) extracted from the requester's free-text request notes, producing a personalized baseline rating instead of defaulting to the global mean.
- Built an interactive NetworkX MultiDiGraph + pyvis visualization of the recommendation network with rating-based color mapping (RdYlGn), aggregated multi-edge statistics, and node-level metrics; performed exploratory analysis on rating patterns across industry groups (academia vs. tech) and reminder-count impacts.
- Deployed the model as a Chrome extension that reads the active recommendation page, sends the requester and recommender IDs to the rating model, and returns an instant predicted response score; the extension also visualizes the local recommendation network and suggests alternative recommenders likely to give stronger ratings.

Autonomous Pet Waste Collection Robot — “Doggie Doo”

Senior Design 2025 – 2026

The University of Texas at Austin

Austin, TX

- Designed, built, and integrated an autonomous outdoor robotic prototype capable of detecting, approaching, collecting, and storing pet waste; applied a systems engineering approach to translate an ambiguous consumer problem into measurable engineering requirements spanning perception, mobility, manipulation, control, and safety.
- Led development of a full robotic architecture composed of a differential-drive mobile platform, vision-based detection pipeline, bucket-and-sweeper collection mechanism, enclosed waste storage system, onboard sensing package, and layered control framework coordinating high-level decision-making with low-level actuation and feedback control.
- Engineered a custom mechanical waste collection system including an articulated bucket-and-sweeper mechanism; performed torque sizing for sweeper and lift mechanisms, drivetrain torque estimation under worst-case conditions, and FEA-backed structural evaluation of critical mounts.
- Designed and validated a geared drivetrain using custom 3D-printed double-helical gears; calculated pitch diameters and center distances for proper meshing; iterated through multiple revisions after physical assembly exposed gear backlash, misalignment, and friction losses not apparent in CAD.
- Supported deployment of a YOLO-based computer vision pipeline for real-time waste detection, comparing model variants using precision, recall, and F1 metrics; contributed to a Python/Arduino layered software architecture with PID closed-loop wheel speed control, encoder feedback, and ultrasonic obstacle detection.

- Designed and implemented a product data model integrating behavioral and academic performance datasets from multiple sources, creating a unified structured foundation that enabled holistic, data-driven decision-making for educators and administrators tracking student outcomes.
- Built interactive dashboards surfacing key student performance metrics and behavioral trends, enabling teachers and administrators to quickly identify at-risk students and prioritize interventions based on data rather than intuition.
- Implemented an AI-powered chatbot using Palantir AIP enabling natural-language interaction with structured product data; iterated on chatbot behavior using real user feedback to improve accuracy and usability.

- Co-built and launched an alumni operations SaaS platform; onboarded 7 paying organizations containing 1,000+ end users and achieved \$1K+ monthly recurring revenue (MRR) through a subscription and implementation-based revenue model; translated user feedback into product features, prioritized roadmap initiatives, and led onboarding and adoption workflows.
- Conducted market research and managed the full growth funnel (outreach → demo → onboarding); implemented product across 15 organizations and prioritized features to support expansion to 50+ organizations.

- Developed a trucking logistics optimization tool using linear programming to minimize total transportation costs and streamline routing efficiency; designed dynamic Excel-based matrices to calculate per-mile costs, trip distances, and truck allocations, reducing empty truck returns and improving overall route efficiency while lowering carbon emissions.

- Developed a prototype smart bike security system integrating an ESP32, Arduino vibration sensor, and camera module with Firebase cloud storage and a React Native mobile app for real-time tamper detection and push notification alerts; designed and 3D printed a custom SolidWorks encasement optimized for durability, weather protection, and ease of use.

LEADERSHIP & ACTIVITIES

- Co-founded MPS Tech and scaled the organization to 75+ paid members in year one; built programming and corporate partnerships and launched consulting-style project teams delivering real client outcomes while providing students hands-on experience in discovery, analysis, and delivery.
- Serving as project manager for a local dental practice growth engagement; scoped and prioritized 3 deliverables including AI workflow automation, SEO content generation, and a website chatbot targeting a 15% increase in new-patient acquisition through improved lead capture and conversion rate optimization.

Additional Activities: Alpha Lambda Mu Fraternity (Nov 2024 – Present) | Texas Pakistani Student Association – Treasurer (Apr 2025 – Present) | American Society of Mechanical Engineers – Member (Aug 2022 – Present) | Texas Convergent – IoT Build Member (Aug 2022 – May 2023) | University Orchestra (Aug 2022 – Apr 2023)

TECHNICAL SKILLS

Engineering Software: SolidWorks, Ansys (FEA & CFD), AutoCAD, MATLAB
Data & Analytics: Excel (Advanced), Spotfire, Tableau, Power BI, Palantir AIP, R / RStudio, SQL
Programming & ML: Python (pandas, NumPy, scikit-learn, Surprise, NetworkX, pyvis), MATLAB, Arduino / C++, React Native; collaborative filtering (SVD), Gradient Boosting, TF-IDF, cross-validation
Business & PM Tools: Salesforce CRM, Confluence, Agile / Kanban, Waterfall / CPM, 8D / FMEA / Six Sigma, Lean Manufacturing, SEM, X-Ray / CSAM Inspection

AWARDS & ADDITIONAL INFORMATION

Samsung Semiconductor Scholar (2024) — Competitive merit-based scholarship awarded to outstanding engineering interns at Samsung Semiconductor’s Austin site.
Work Authorization: U.S. Citizen